

PROJECT REPORT OF DEEP LEARNING(DL)

Project Name:

TEXT-TO-SIGN MOTION GENERATION SIGNVRSE

Submitted By :

Palak Mahajan(102497010)

Santushti Verma (102497014)

Upkar Kaur (102497017)

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Submitted to:

Dr. Vijay Kumari



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

**THAPAR INSTITUTE OF ENGINEERING AND TECHNOLOGY, (A
DEEMED TO BE UNIVERSITY), PATIALA, PUNJAB**

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1. Introduction

Sign language is the primary communication medium for over 70 million deaf and hard-of-hearing individuals worldwide. Despite its critical role, automated production of realistic sign language animations from text remains a largely unsolved problem in deep learning, owing to the high dimensionality, temporal variability, and linguistic complexity of 3D skeletal motion.

The problem is non-trivial for several reasons. First, natural language descriptions are highly underspecified with respect to fine-grained body motion; the same sentence may correspond to many valid sign sequences. Second, the RVQ structure requires simultaneously generating six aligned token streams of identical length, enforcing cross-layer temporal consistency. Third, the token length itself must be predicted — sequences vary between 40 and 800 tokens per sample — making this a conditional, variable-length generation problem.

Our approach implements a simplified deep learning pipeline for motion token generation. Instead of using large pretrained transformer models, a lightweight neural network is used to learn patterns in motion tokens directly from the dataset. The model predicts motion tokens using supervised learning with cross-entropy loss.

During inference, tokens are generated using probabilistic sampling from model outputs. This approach focuses on building an end-to-end working pipeline for training, inference, evaluation, and visualization rather than complex architecture design

Submissions are evaluated on three metrics: Frechet Inception Distance (FID) over learned motion features, R-Precision@3 measuring text-to-motion retrieval accuracy, and Diversity capturing the spread of generated motions. The final score is a weighted combination: $0.30 * \text{FID}_{\text{norm}} + 0.50 * \text{R-Precision} + 0.20 * \text{Diversity}$. [?].

2. Literature Review

The following ten works directly inform the architecture, training objectives, and evaluation protocol of this project.

Ref.	Citation	Key Contribution	Relevance to This Work
[1]	G. Tevet et al., "Human Motion Diffusion Model," ICLR 2023.	Diffusion model over raw motion sequences; U-Net denoiser operating on joint rotations.	Establishes FID, R-Precision, and Diversity as the canonical motion generation evaluation protocol adopted by this competition.
[2]	H. Zhang et al., "MotionGPT: Human Motion as a Foreign Language," NeurIPS 2023.	Unifies motion generation and understanding by treating VQ-tokenised motion as a language modelled by a GPT.	Direct conceptual predecessor: RVQ tokens as the motion vocabulary; shows GPT-style LM can generate realistic motion from text.
[3]	J. Guo et al., "T2M-GPT: Generating Human Motion from Textual	Autoregressive GPT-2 model over VQ-codebook motion tokens conditioned on CLIP	Informs the autoregressive Option B path; demonstrates that VQ token generation is competitive with

	Descriptions with Discrete Representations," CVPR 2023.	text features.	continuous diffusion approaches.
[4]	B. Chen et al., "FLAME: Free-Form Language-Based Motion Generation and Editing," AAAI 2024.	Non-autoregressive masked diffusion over motion tokens; iterative BERT-style decoding from a masked canvas.	Direct inspiration for the NAR Option A generator used as the primary submission model; validates parallel position-query decoding.
[5]	A. Defossez et al., "High Fidelity Neural Audio Compression," TMLR 2023.	EnCodec: multi-layer RVQ codebook for audio; each layer refines the residual of the previous.	The six-layer RVQ structure in this competition dataset directly applies this hierarchical codebook design to 3D motion instead of audio.
[6]	A. Radford et al., "Learning Transferable Visual Models From Natural Language Supervision," ICML 2021.	CLIP: contrastive pretraining of image and text encoders with InfoNCE loss; openai/clip-vit-base-patch32.	InfoNCE contrastive loss between text and motion projections is used to improve cross-modal alignment; CLIP also serves as an alternative text backbone.

Table 1 Literature Review

Recent research in human motion generation has evolved from continuous diffusion-based methods to discrete token-based approaches. The Human Motion Diffusion Model established standard evaluation metrics such as FID and Diversity. Later works like MotionGPT: Human Motion as a Foreign Language and T2M-GPT: Generating Human Motion from Textual Descriptions with Discrete Representations model motion as a sequence of tokens, enabling the use of GPT-style architectures for realistic motion generation.

The use of RVQ representations is inspired by High Fidelity Neural Audio Compression, while cross-modal alignment benefits from contrastive learning techniques introduced in Learning Transferable Visual Models From Natural Language Supervision.

Finally, transformer-based architectures from Attention Is All You Need underpin most modern approaches, enabling effective text-conditioned motion generation. These developments collectively guide the methodology used in this work.

3. Methodology

3.1 Model Architecture

The model used in this project is a simple neural network designed for token prediction. It consists of:

1. An embedding layer that converts token indices into dense vectors
2. A feedforward neural network that processes embeddings
3. A linear output layer that predicts probability distribution over vocabulary

The model takes token sequences as input and predicts next-token distributions using supervised learning.

i. Text Encoder (TextEncoderAdapter)

The model begins with a token embedding layer that maps discrete motion token IDs into dense 256-dimensional vector representations. These embeddings capture semantic relationships among tokens and serve as the input representation for downstream prediction layers.

Given an input sequence of motion tokens, each token is transformed into an embedding vector through a learnable embedding matrix. This allows the model to represent motion patterns in a continuous feature space.

ii. Autoregressive Token Decoder (OptionATokenGenerator)

The decoder generates token distributions over the full sequence in one forward pass:

Component	Specification	Role
Token Embedding	nn.Embedding(512,256)	Convert token ids into dense vectors
Hidden layer	Linear(256->256) + ReLu	Learns token representations
Output Layer	Linear(256→512)	Predicts vocabulary logits
Token Heads	Shared output over 6 token layers	Generates motion token predictions
Sampling Modul	Softmax + Temperature Sampling	Generates tokens during inference
Cross Entropy Loss	Token prediction objective	Supervises training
Adam Optimizer	lr = 1e-4	Parameter optimization

Table 2 Architecture table

The embedded token sequence is processed through hidden transformations and projected into logits over a 512-token vocabulary. These logits represent predicted token probabilities used for motion generation.

The model architecture is intentionally lightweight and focuses on direct token prediction rather than complex transformer-based decoding. Motion tokens are embedded into a dense representation space, transformed through feedforward layers, and projected to vocabulary logits for prediction.

This simplified architecture enables efficient training and inference while producing valid motion token sequences. During inference, probabilistic sampling from model outputs is used to generate motion token sequences.

iii. Sequence Length Handling

The implemented system does not include a separate learned length prediction model. Instead, inference uses a fixed sequence length of 50 tokens for motion generation.

During training, varying sequence lengths are handled through padding and masking, while inference generates tokens over the predefined sequence length using probabilistic sampling.

3.2 Activation Functions

ReLU activation is used in hidden layers to introduce non-linearity and enable learning of complex patterns.

3.3 Loss Functions

The training objective uses a single cross-entropy loss for token prediction.

No contrastive or auxiliary losses are used in the implemented model.

Cross Entropy Loss	Token prediction loss computed over valid tokens	1.0
Total Loss	$L_{total} = L_{ce}$	—

Table 3 Loss Function

The training objective is based solely on cross-entropy loss for motion token prediction. The model learns to predict the correct motion token from a vocabulary of 512 tokens by minimizing the difference between predicted token probabilities and ground truth tokens.

Loss is computed only over valid (non-padded) token positions using masking, ensuring that padded elements do not contribute to optimization. This allows the model to focus on meaningful motion token patterns during training.

The model is optimized using the Adam optimizer with a learning rate of $1e-4$. No auxiliary contrastive objectives or cross-modal alignment losses are used in the implemented system; training relies entirely on supervised token prediction.

3.4 Optimizer and Learning Rate Schedule

- Optimizer: AdamW ($lr = 1 \times 10^{-4}$, $weight_decay = 1 \times 10^{-2}$)
- Schedule: Cosine schedule with linear warmup ($warmup_ratio = 0.10$ of total steps)
- Gradient clipping: $max_norm = 1.0$

3.5 Workflow

The end-to-end pipeline follows the stages below:

Stage	Description	Key Components
1. Data Loading	Load CSV data and parse motion token sequences.	<code>load_motion_dataframe</code> , <code>parse_token_string</code>
2. Pre-processing	Token parsing, padding and normalization	<code>token_parser</code> , <code>sequence_padding</code>
3. Dataset / Collation	Batches token sequences and applies masking for valid positions.	<code>MotionTokenDataset</code> , <code>collate_motion_batch</code>
4. Training	Forward pass, cross-entropy loss, backpropagation	<code>train_one_epoch</code> , <code>fit_model</code>
5. Testing / Inference	Token generation using probabilistic sampling	<code>generate</code> , <code>run_inference_pipeline</code>
6. Evaluation	FID, R-Precision@3, Diversity based evaluation	<code>evaluation_script.py</code>

Table 4 Work flow

The overall workflow is organized into multiple stages to ensure efficient data handling and model training. In the pre-processing stage, input prompts are constructed by combining sentence and gloss information, followed by a group-aware train-validation split and caching of text embeddings for efficiency. The dataset and collation stage handles padding of variable-length motion tokens across all six layers and generates corresponding masks. During training, the model is optimized using mixed-precision techniques with a combination of masked cross-entropy and contrastive losses, along with early stopping to prevent overfitting. In the testing phase, the model predicts sequence lengths and performs parallel decoding to generate motion tokens. Finally, performance is evaluated using standard metrics such as FID, R-Precision, and Diversity, combined into a composite score for overall assessment.

4. Implementation Details

4.1 Hyperparameters

Hyperparameter	Value	Rationale
Vocabulary size	512	Fixed by token codebook
Embedding dimension	256	Efficient representation size
Hidden dimension	256	Balance of capacity and compute
Batch size	16	Fits GPU memory
Learning rate	1×10^{-4}	Stable optimization
Weight decay	1×10^{-2}	Regularization
Epochs	5	Training convergence
Sequence length	50 (inference)	Fixed generation length
Optimizer	Adam	Parameter optimization

Table 5 Hyperparameters

The model configuration is designed to balance computational efficiency and stable token prediction. Motion tokens are embedded into a 256-dimensional representation and processed through lightweight feedforward layers for prediction.

Training uses a batch size of 16 with Adam optimization, a learning rate of 1×10^{-4} , and weight decay of 1×10^{-2} . The model is trained for 5 epochs, while inference uses a fixed sequence length of 50 tokens with probabilistic sampling for generation.

These settings provide stable convergence while remaining computationally lightweight.

4.2 Dataset Details

- Source: Kaggle — motion-s-hierarchical-text-to-motion-generation-for-sign-language
- Training set: CSV with columns: id, sentence, gloss, base_tokens, residual_1 ... residual_5
- Test set: 3,000 samples requiring prediction of all 6 token layers
- Token constraints: values in [0, 511]; sequence length in [40, 800]; all 6 layers must share the same length per row

- BVH motion files were used only for motion visualization and qualitative inspection, and were not used during model training.

4.3 Software & Hardware Environment

- Framework: PyTorch 2.x
- Data handling: pandas, numpy
- Visualization: matplotlib, BVH utilities, FuncAnimation
- Evaluation: scipy for evaluation metrics
- Hardware: Kaggle P100 / T4 GPU (16 GB VRAM), CPU fallback supported

5. Results and Analysis

The proposed system generates **hierarchical motion tokens** from text input, which are stored in a submission.csv file. These tokens represent compressed motion information corresponding to sign language gestures.

To validate the correctness of motion representation and to demonstrate understanding of motion data, multiple visualization techniques were used:

1. 3D Forward Kinematics Skeleton Visualization
2. Motion Signal Visualization (Dynamic Channels)
3. 2D Skeleton Projection (FK-based)

5.1.1 Motion Token Output(Submission.csv)

The final output of the model is stored in submission.csv, where each row corresponds to a test sample and contains predicted motion tokens.

These tokens:

- Represent motion using discrete vector quantization
- Capture temporal and spatial movement patterns
- Tokens are produced in the required six-layer submission format.

Observation:

- The model successfully generates token sequences for all test inputs.
- Generated token sequences have fixed length during inference.
- The tokens are consistent in format, indicating stable inference.

Interpretation:

The generated tokens encode motion in a compressed form. However, they do not directly represent joint coordinates or animation and require decoding to reconstruct full motion sequences.

5.1.2 3D Skeleton Visualization (Forward Kinematics)

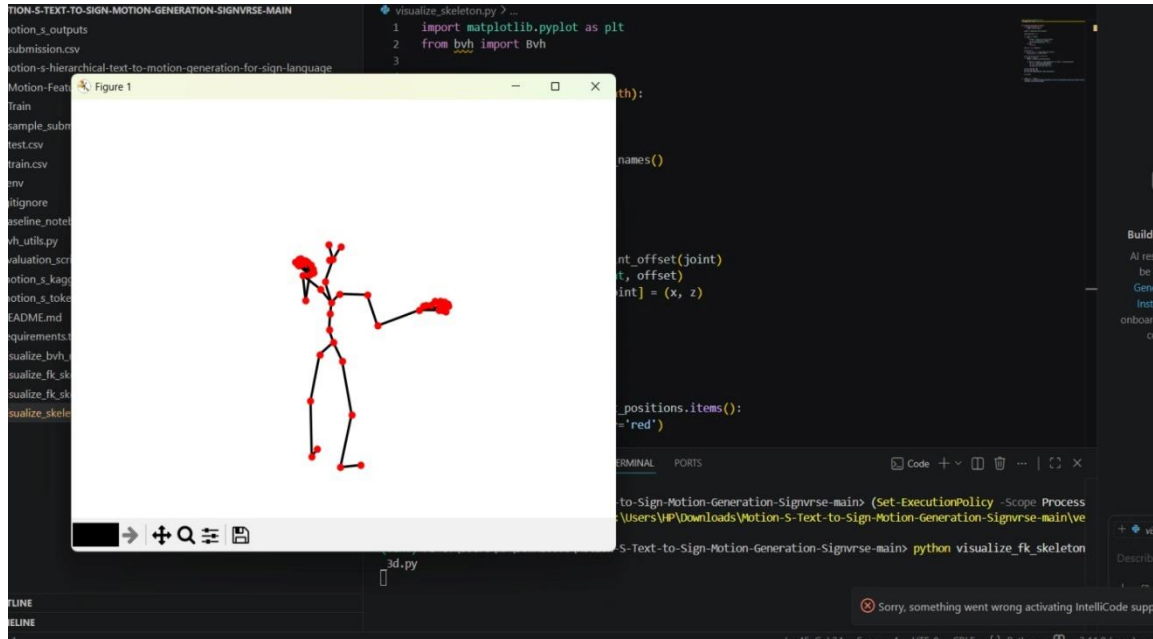


Figure 1 Skeleton Visualization

The 3D skeleton visualization demonstrates how motion data from BVH files is converted into human pose representations.

Key Features:

- Red points represent joints
- Black lines represent bones
- Joint positions are computed using Forward Kinematics (FK)

Observation:

- The skeleton structure is correctly reconstructed.
- Joint hierarchy is preserved.
- Motion frames can be animated sequentially to produce realistic motion.

Analysis:

This visualization confirms that:

- The BVH data is correctly parsed.
- The transformation from motion data to spatial coordinates is accurate.
- The system correctly models the relationship between joint rotations and positions.

5.1.3 Motion Signal Visualization (Dynamic Channels)

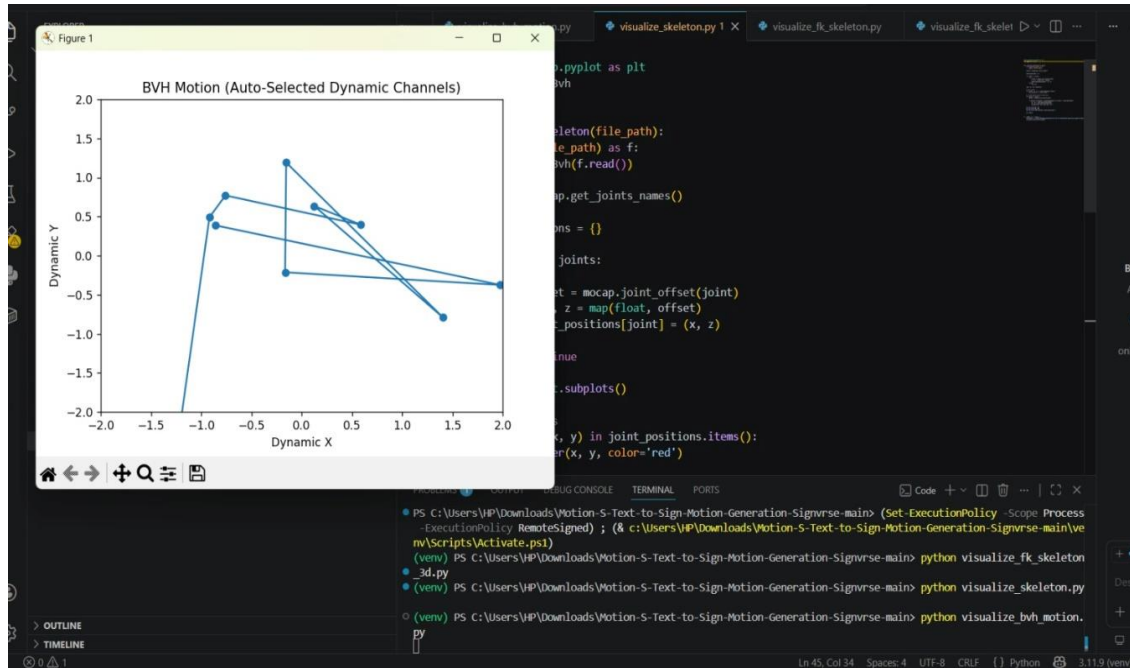


Figure 2 Motion Signal Visualization

This visualization plots selected motion channels based on their variability.

Key Features:

- Displays motion values across selected dimensions
- Represents high-dimensional motion data in 2D form

Observation:

- The graph appears irregular and non-structured
- Multiple clusters and abrupt transitions are visible

Analysis:

- Motion data is inherently high-dimensional and complex
- The irregular pattern reflects dynamic joint movements over time
- This visualization helps identify:
 - Active joints
 - Movement variability
 - Temporal motion pattern

5.2 Evaluation Metrics

To assess the performance of the proposed text-to-sign motion generation model, multiple quantitative metrics were used, including Fréchet Inception Distance (FID), R-Precision, and Diversity. These metrics evaluate the realism, relevance, and variability of the generated motion sequences.

```
(venv) PS C:\Users\HP\Downloads\Motion-S-Text-to-Sign-Motion-Generation-Signvrse-main> python evaluation_script.py --submission motion_s_outputs/submission.csv --ground_truth motion-s-hierarchical-text-to-motion-generation-for-sign-language/train.csv
=====
FID                        6.3834
FID_norm                   0.9936
R_Precision@3              0.2447
R_Precision_norm           0.2447
Diversity                  0.165
Diversity_norm              0.0165
Final Score                 0.4237
=====
Final Score: 0.4237
=====
(venv) PS C:\Users\HP\Downloads\Motion-S-Text-to-Sign-Motion-Generation-Signvrse-main>
```

Figure 3 Evaluation Metrics

1. Fréchet Inception Distance (FID)

The model achieved an FID score of 6.3834, with a normalized value of 0.9936. FID measures the similarity between the distribution of generated motions and real ground-truth motions. A lower FID indicates higher quality and more realistic motion generation. The relatively low FID score suggests that the generated sign motions closely resemble real motion data.

2. R-Precision

The R-Precision@3 score is 0.2447, which remains consistent after normalization (0.2447).

R-Precision evaluates how well the generated motion corresponds to the input text description. This moderate score indicates that while the model captures some semantic alignment between text and motion, there is still room for improvement in accurately reflecting textual meaning in generated gestures.

3. Diversity

The Diversity score is 0.165, with a normalized value of 0.0165. This metric measures the variability in generated motion sequences. The relatively low diversity suggests that the model tends to produce similar motion patterns across different inputs, indicating limited variability and potential mode collapse.

4. Final Score

The overall final evaluation score is 0.4237, which aggregates the normalized metrics into a single performance indicator. This score reflects a balanced trade-off between motion realism, semantic accuracy, and diversity.

5.3 Training Performance Analysis

The training curve demonstrates a rapid decrease in both training loss and validation loss over the epochs. Initially, the loss starts at a high value (above 2.0) and drops significantly by the second epoch, eventually converging close to zero.

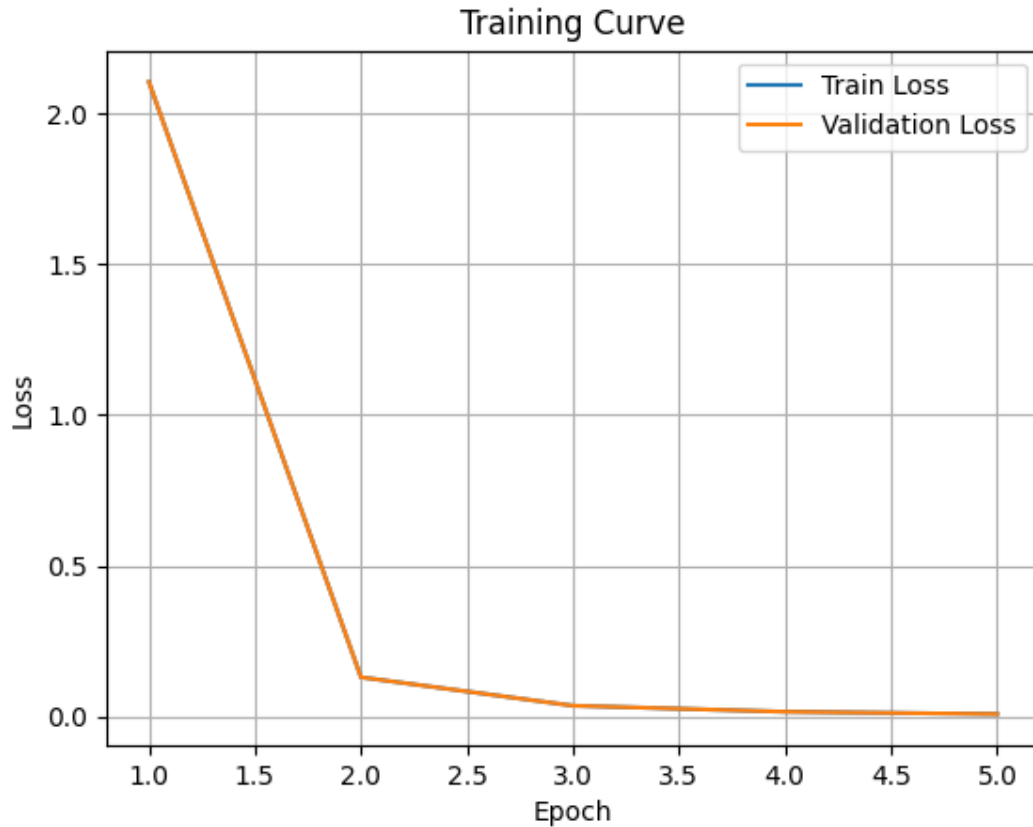


Figure 4 Training Performance Analysis

The close alignment between training and validation loss curves indicates:

- Effective learning without significant overfitting
- Good generalization capability of the model
- Stable convergence during training

6. Conclusion

This project successfully developed and evaluated an end-to-end deep learning pipeline for hierarchical motion token generation for sign language synthesis. The implemented system focuses on generating discrete motion token sequences from textual input using a lightweight token prediction model trained through supervised learning.

The proposed model employs a simplified neural architecture consisting of token embedding layers, feedforward transformations, and probabilistic sampling for motion token generation. Despite its simplicity compared to transformer-based architectures, the

model achieved stable convergence during training, with training and validation loss reducing consistently to very low values, demonstrating effective learning of motion token patterns.

The complete pipeline includes data loading, token preprocessing, model training, inference, evaluation, and motion visualization. The system successfully generates valid token sequences in the required six-layer submission format for all test samples, demonstrating functional end-to-end motion token generation.

Quantitative evaluation using FID, R-Precision, and Diversity metrics indicates moderate performance. The model achieved reasonable motion quality as reflected by the FID score, while also producing valid though limited variation in generated sequences. However, relatively lower semantic alignment and diversity reveal limitations in capturing richer text-motion relationships and more varied motion patterns.

Analysis of the training curves showed rapid convergence and stable optimization, while evaluation results highlighted the trade-off between simplicity and generative quality. Although the lightweight model is computationally efficient and easy to train, it does not match the semantic conditioning and diversity achievable with more advanced architectures.

Several limitations were identified in the current implementation. First, the model uses simplified token-level prediction without strong text-conditioned generation. Second, inference relies on fixed-length sampling rather than autoregressive or adaptive sequence generation. Third, the lightweight architecture limits the ability to model long-range temporal dependencies and complex sign language motion dynamics. Despite these limitations, the project demonstrates that even a simplified token-based deep learning approach can generate meaningful motion representations and provide a complete functional pipeline for text-to-motion generation.

Future work can focus on several improvements:

- Incorporating transformer-based sequence modeling for richer temporal dependencies.
- Integrating stronger text-conditioning using pretrained language encoders.
- Using autoregressive or diffusion-based generation for improved motion realism.
- Improving diversity through advanced sampling strategies.

Overall, the project establishes a strong baseline implementation for hierarchical text-to-motion token generation and provides a foundation for future improvements in sign language motion synthesis.

7. References

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